

EREN KARL T. MORTEL

2026

Home Address:

B.4 L. 30 Cedar Street
Brgy. Bagong Silangan, Quezon City
Metro Manila, Philippines 1191

Home Phone: 441-2139

Cell Phone: +63 976-019-8220

Home e-mail: morteleren@gmail.com

LinkedIn Profile: [linkedin.com/in/erenmortel](https://www.linkedin.com/in/erenmortel)

PUP Address:

College of Engineering and Architecture
Anonas St., Sta. Mesa
Metro Manila, Philippines

E-mail:

erenkarlmortel@iskolarnghayan.pup.edu.ph

Education

- B.S. 2026, Polytechnic University of the Philippines, Railway Engineering and Management. Thesis: "Modelling Pedestrian Flow and Analyzing Emergency Egress Scenarios in Transit Stations: A Case Study for the MRT-3 Ayala Station."
- S.H.S 2021, University of the Cordilleras, Science, Technology, Engineering and Mathematics. Capstone: "The Physics of Athletic Throwing Events," With High Honors.
- J.H.S 2019, New Era University, With High Honors, Best in Drafting.

Current Positions and Affiliations

- Volunteer, Philippine Animal Welfare Society (PAWS), 2025–present.

Previous Positions

Data Processing Intern (City Planning and Dev't Dept.), Quezon City Government, February 2026–May 2026.
Cadet Design Engineer, STECONFER Phil Corp., July 2025–September 2025.
Creative Head, PUP – Railway Engineering Students' Society, September 2024–September 2025.
Graphic Designer, HPL Gamedesign Corp., March 2022–October 2022.

Certifications

Civil Service Eligible (Professional), 2025.

Honors and Awards – PUP

- Best Research Proposal (Pre-oral Defense), 2025.
Given alongside Elmon Bandoja and Andrea Rivera: "In recognition of an exemplary research framework [and] exceptional command of the subject matter [...] that distinguished the recipient's work as the best research proposal during the defense."
- Best Presenter (Pre-oral Defense), 2025
"In recognition of outstanding presentation skills [and] exceptional command that distinguished the recipient as the most commendable presenter of the final defense."
- President's Lister, 2022–2023. Recognized for maintaining a GWA above 1.50.
- Dean's Lister, 2022. Recognized for maintaining a GWA above 1.75.

Honors and Awards – UC

- Senior High Graduation, High Honors, 2021.
Given every moving up celebration to recognize the student who has reached a high distinction within their

class.

Honors and Awards – NEU

Moving up, Best in Drafting, 2019.

Given every moving up celebration to recognize the student who has reached the highest general weighted average (GWA) in one particular elective.

Moving up, High Honors, 2019.

Given every moving up celebration to recognize the student who has reached a high distinction within their class.

Technical Reports and Projects

Lead Developer, Las Piñas-Zapote Riverbank Active Mobility Trail, June 2025. Developed a 186-km ecological trail plan using GIS and SketchUp; presented to a panel of urban planning professionals.

Technical Contributor, MRT-7 Northern Line Construction Plans, 2025. Produced rail welding plans and plinth reinforcement layouts; conducted site surveys to verify design-to-field alignment.

Presentations

Presenter, “Modelling Pedestrian Flow and Analyzing Emergency Egress Scenarios in Transit Stations: A Case Study for the MRT-3 Ayala Station.” Railway Academe – Industry Linkage (RAIL), 3rd Research Advancement Forum, Philippine Railways Institute (PRI) and Department of Transportation (DOTr), hosted by the UP National Center for Transportation Studies (UP NCTS), Diliman, Quezon City, June 2026. Certificate of Participation issued by Acting Director Anneli R. Lontoc, CESO I, PRI, and the Undersecretary for DOTr. Event attended by Engr. Deovanni S. Miranda, General Manager, Philippine National Railways (PNR).

Workshops Conducted

Creative Head for “Track to the Future 5” at Polytechnic University of the Philippines – College of Engineering and Architecture, Audio-Visual Room, June 30, 2025. Over 100 participants from all year levels of the program; including alumni and senior faculty. Developed for the Railway Engineering Students’ Society.

Creative Head for “The 2nd Philippine Railway Engineering Student Summit (PRESS 2025)” at Polytechnic University of the Philippines – College of Communication Theater, June 10, 2025. Coordinated with industry partners including PPI Pazifik Power, Inc., RC Cola, and Microcadd Institute, Inc., as well as academic societies from FEATI University and Technological University of the Philippines. Two separate 4-hour sessions; over 150 distinguished participants and speakers, including senior engineers and managers from global firms Colas Rail and Alstom. Developed for the Railway Engineering Students’ Society.

Creative Head for “RE Fest: Carnival Clash: Rhythm of Revelry” at Polytechnic University of the Philippines – College of Engineering and Architecture, April 22, 2024–May 2, 2024. Over 100 participants from all year levels of the program. Developed for the Railway Engineering Students’ Society.

Developer and co-facilitator (with Michael Vince Mapula) of “The Art of Personal Branding: Building Your Identity as a Railway Engineer” at Polytechnic University of the Philippines – College of Engineering and Architecture, Audio-Visual Room, February 24, 2025. Two separate 1-hour sessions; over 100 participants, including senior leadership. Developed for the Railway Engineering Students’ Society.

Creative Head for “REMastering Research: Empowering Scholars for Academic Excellence” at Polytechnic University of the Philippines – Claro M. Recto Hall, January 18, 2025. Two separate 1-hour sessions; over 100 participants, including senior researchers, previous alumni, and faculty. Developed for the Railway Engineering Students’ Society.

Creative Head for “Advanced Tools for Fast-Track Data and Workflow Management” at Polytechnic University of the Philippines – College of Communication, Audio-Visual Room, December 12, 2024. Developed for the Railway Engineering Students’ Society.

Creative Head for “International Conference and World Café: Advancing Transportation and Construction Management in the Philippines and the United Kingdom” at Polytechnic University of the Philippines – Claro M. Recto Hall, November 22, 2024. Coordinated with the British Council, the University of the West of England (UWE) Bristol, the PUP Aggregates, the Philippine Institute of Civil Engineers, and the PUP Office of International Affairs to design and execute high-profile promotional assets. Two separate 5-hour sessions; over 100 distinguished participants and speakers, including the University President, Dr. Manuel M. Muhi, and Head of the Rail Sector of First Balfour, Antony Di Rosa.

Creative Head for “Leadership Training Program” at Polytechnic University of the Philippines – College of Engineering and Architecture, Audio-Visual Room, October 1, 2024. Developed for the Railway Engineering Students’ Society.

Professional Associations and Service

Railway Engineering Students’ Society
Philippine Animal Welfare Society

Areas of Research Interest

Quantitative modeling of human movement and spatial performance in high-density transit infrastructure, specifically the assessment of pedestrian Level of Service (LOS) and the optimization of emergency egress efficiency to ensure public safety in urban railway systems.

Integration of active mobility corridors and nature-based solutions within flood-prone metropolitan regions, focusing on the revitalization of riverine ecosystems as both social infrastructure and critical components of urban climate resilience.

The intersection of large-scale transportation engineering and sustainable land-use planning, with an emphasis on Transit-Oriented Development (TOD) frameworks and the institutional coordination required to align national infrastructure projects with local development goals.

- Application of digital visualization and simulation tools in urban design, aiming to bridge the gap between technical engineering specifications and human-centric planning to create inclusive, accessible, and legible public spaces.

References Available upon request.